CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Aachen • EU Blue Card linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in machine learning and data engineering, from data pipelines, ETL and feature engineering to training PyTorch and TensorFlow models, with Docker-based cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in time-series forecasting, SQL, NoSQL and vector databases.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full data pipeline for knowledge extraction end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and optimized multi-agent extraction algorithms with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, validated against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, Postgres, MongoDB (NoSQL), Neo4j, Vector Search, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an AI agent with LangGraph that routes each request across vector database queries, general LLM inference and summarization, after mapping the decision-making workflows of business stakeholders.
- Developed a React front-end and a Python REST API backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Designed Retrieval-Augmented Generation (RAG) pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, CDK), Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Cell KPI Anomaly Pipeline | Kafka | PySpark | PyTorch | REST/gRPC | Docker

- Data pipeline for synthetic 5G cell performance counters: Kafka producer and consumer, ingestion via REST and gRPC endpoints, PySpark ETL with feature engineering, data quality and validation checks, features stored in Postgres.
- Anomaly detection with an Isolation Forest baseline vs. a small PyTorch autoencoder, model monitoring of prediction drift, Docker Compose setup and pytest in GitHub Actions.

TECHNICAL SKILLSET

- **Programming:** Python (Pandas, NumPy, SciPy), SQL
- **Machine Learning:** PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost, LSTM, time-series forecasting (ARIMA, SARIMA, Prophet), feature engineering
- **Data & Databases:** data pipelines, ETL, Postgres, MongoDB (NoSQL), Neo4j, Vector Databases (Qdrant, Pinecone, Azure AI Search), Streamlit
- **GenAI & NLP:** LangGraph, LangChain, LiteLLM, RAG, Hugging Face Transformers, NLTK, RAG evaluation (RAGAS, PytrecEval)
- **MLOps & Cloud:** Git, GitLab CI/CD, Docker, Kubernetes, FastAPI, AWS (Lambda, SQS, DynamoDB, EC2, CDK), Azure
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT